

EDV28A-N-01 系列产品使用说明书

EDV28A-N-01 Series User Manual



深圳市易探科技有限公司

Shenzhen EasyDetek Technology Co., Ltd.

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V1.0.0	2026-07-06	研发部 Development Department	初版 Initial version
V1.0.1	2026-07-22	研发部 Development Department	更新指示灯状态及功能说明图片 Update LED status and functional description images
V1.0.2	2026-07-27	研发部 Development Department	新增目录、产品详细介绍、使用指南、通信协议规范、保修条款及中英文对照 Add directory, product introduction, user guide, communication protocol specification, warranty terms, and English and Chinese translation
V1.0.3	2026-08-06	研发部 Development Department	新增小程序端说明 Add Mini Program instructions

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第一部分 / Chapter 1

产品介绍 / Product Introduction



图 1.1: 产品展示图
Fig1.1: Product View

1.1 产品概述 / Overview

EDV28A-N-01 是一款基于 60GHz 毫米波雷达技术的在位体征监测传感器，可实时探测胸腔毫米级微动，提供舒适无感非接触式长期稳定监测。产品结合边缘计算和智能算法，连续监测分析目标在位情况，及心率、呼吸、体动等关键指标。

The EDV28A-N-01 is an in-place vital signs monitoring sensor based on 60GHz millimeter-wave radar technology. It can detect millimeter-level chest movements in real time, providing comfortable, non-contact long-term stable monitoring. Combining edge computing and intelligent algorithms, it continuously monitors and analyzes the target's presence status as well as key indicators such as heart rate, respiration, and body movement.

1.2 产品特点 / Features

- 非接触监测：采用前沿 60G 毫米波传感技术，实时探测胸腔毫米级微动，提供舒适无感非接触式长期稳定监测。

Non-contact Monitoring: Utilizing cutting-edge 60GHz millimeter-wave sensing technology to detect millimeter-level chest movements in real time, providing comfortable, non-contact long-term stable monitoring.

- 智能分析：结合边缘计算和智能算法，连续监测分析目标在位情况，及心率、呼吸、体动等关键指标。
- Intelligent Analysis: Combining edge computing and intelligent algorithms to continuously monitor and analyze the target's presence status, heart rate, respiration, and body movement.

- 睡眠监测：智能监测睡眠阶段的浅睡、深睡、REM 睡眠、清醒期等情况，可评估和复盘睡眠质量。
- Sleep Monitoring: Intelligently monitors sleep stages including light sleep, deep sleep, REM sleep, and

wakefulness, enabling sleep quality assessment and review.

- 异常警报：智能识别异常呼吸或心率，支持及时警报，守护用户健康安全。
Abnormal Alert: Intelligently identifies abnormal respiration or heart rate, supports timely alerts to safeguard user health and safety.
- 隐私安全：无摄像隐私风险，在物理层和数据层完全严格遵守隐私法规，保护用户信息安全。
Privacy Security: No camera privacy risks, fully complies with privacy regulations at both physical and data layers to protect user information security.

1.3 应用场景 / Applications

- 商务办公：安装在办公室区域，实现人员在位监测，智能管理办公环境能耗与安全。
Business Office: Installed in office areas for occupancy monitoring, enabling intelligent management of office energy consumption and safety.
- 居家养老：安装在卧室或客厅，实时监测老人心率、呼吸和体动，异常情况及时警报。
Home Care: Installed in bedrooms or living rooms to monitor the elderly's heart rate, respiration and body movement in real time, with timely alerts for abnormalities.
- 商业场所：安装在酒店客房或公寓，实现入住检测与智能客房管理。
Commercial Places: Installed in hotel rooms or apartments for occupancy detection and smart room management.
- 学校宿舍：安装在宿舍区域，监测学生作息情况，保障住宿安全。
School Dormitories: Installed in dormitory areas to monitor students' daily routines and ensure accommodation safety.
- 月子中心：安装在母婴房间，监测产妇与婴儿体征，守护母婴健康。
Postpartum Care Centers: Installed in maternal rooms to monitor vital signs of mothers and infants, safeguarding their health.
- 康养机构：安装在养老院或康复中心，24 小时连续监测长者体征，提升照护效率。
Wellness Institutions: Installed in nursing homes or rehabilitation centers for 24/7 continuous vital signs monitoring, improving care efficiency.

1.4 外形结构 / Appearance Structure

1.4.1 外形尺寸 / Dimension

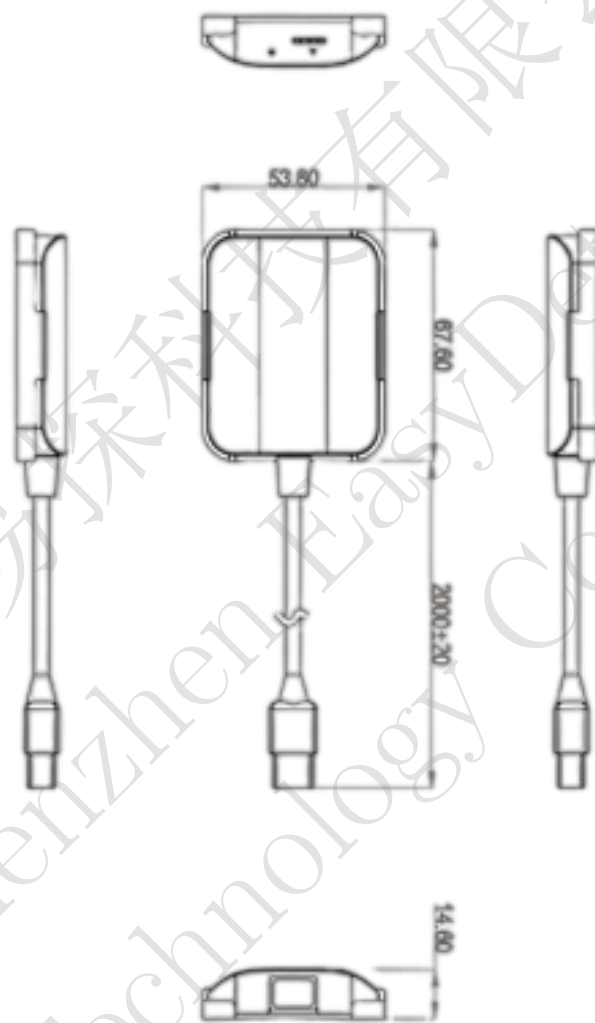


图 1.2: 产品外形尺寸图
Fig1.2: Product Outline Dimension Drawing

表 1.1: 外形尺寸参数
Tab1.1: Dimension Parameters

参数名称 Parameter	数值 Value	单位 Unit
主体长度 Overall Length	67.60	mm
主体宽度 Overall Width	53.80	mm
主体厚度 Overall Height	14.60	mm
连接线总长 Cable Length	2000±20	mm

1.4.2 接口说明 / Electrical Interface

表 1.2: 接口功能表
Tab1.2: Interface Function Table

接口序号 Number	名称 Name	功能 Function
1	VCC	5V 供电电源, 使用 5V2A USB Type-A 接口的适配器进行供电。 5V Power Supply, use a 5V2A USB Type-A adapter for power supply.

1.5 性能参数 / Parameters & Performances

1.5.1 参数列表 / Parameters

表 1.3: 参数表
Tab1.3: Parameter Table

类别 Category	参数 Parameter	可配置? Config.?	数值 Value	说明 Remarks
发射参数RF Parameters	工作频段 Operating Freq.	N	59~64 GHz	毫米波雷达频段 Millimeter-wave radar band
	最大发射功率 Max TX Power	N	≤15 dBm	雷达最大发射功率 Max radar TX power
性能参数Specifications	探测距离 Detection Distance	Y	<1.5 m (默认 0.7 m/ Default 0.7 m)	可配置探测范围 Configurable detection range
	在位检测时间 Presence Detect Time	N	<5 s	目标出现在区域内到检测到的时间 Time to detect presence

续下一页

表 1.3: 参数表 (续)
Tab1.3: Parameter Table(continued)

类别 Category	参数 Parameter	可配置? Config.?	数值 Value	说明 Remarks
	离位检测时间 Absence Detect Time	N	5~100 s	目标离开到检测为离位的时间 Time to detect absence
	呼吸率范围 Respiration Rate	N	12~40 bpm	呼吸率测量范围 Respiration rate measurement range
	心率范围 Heart Rate	N	60~120 bpm	心率测量范围 Heart rate measurement range
	数据上报周期 Data Report Period	Y	1~60 s (默认 3 s/ Default 3 s)	数据上报间隔, 可配置 Data upload interval, configurable
电气参数Electrical	工作电压 Operating Voltage	N	5 V	直流供电电压 DC power supply voltage
	平均功耗 Average Power	N	<0.4 W	正常工作平均功耗 Average power consumption
	最大功耗 Max Power	N	<1 W	峰值功耗 Peak power consumption
	通讯方式 Communication	N	MQTT	设备和客户端通信协议 Network communication protocol
运行参数Operating Params	工作温度 Operating Temp	N	-20~60°C	正常工作环境温度范围 Normal operating temp range
	存储温度 Storage Temp	N	-20~85°C	存储环境温度范围 Storage temperature range

* 注：“可配置？”指的是是否提供接口供用户根据需要对该参数在允许范围内进行配置。

***Note:** "Configurable?" means whether an interface is provided for users to set the parameter within the permissible range as required.

深圳市易探科技有限公司
Shenzhen EasyDetek
Technology Co., Ltd.

第二部分 / Chapter 2

命名规范 / Naming Rules

制造商 Manufacture	频段 Frequency	产品大类别 Parent	产品细分类别 Child	产品编号 Part No.	功能扩展 Extended Functions	流水号 Serial No.
ED	V	2	8	A	N	01
易探 EasyDetek	C 5.8GHz	1 雷达模组 RD Module	0 超低功耗 low power consumption	0 ~ 9 A ~ Z	Y 有光敏 With Photosensitive	A Matter 1 Cat-1
	X 10.5GHz	2 独立传感器 Independent Sensor	1 旗舰 flagship		N 无光敏 Without Photosensitive	C Casambi 2 2.4G
	Q 24GHz	3 天线 & 器件 Antennas & Devices	2 侧装 Side-mounted		G 干接点 Dry node	D Dali 4 RS485
	V 60GHz	4 MCU	3 外置可调 External Adjustable		S AC 开关 AC Switch	K KNX B BLE
	S 生态产品 Eco-products	5 配件 Accessories	5 通用 General			M Mi P PLC
		6 IC	6 户外 Outdoor			T Tuya R RF433
		7 AI 传感 AI Sensor	7 保留 Reserved			W Wifi
		8 组网 Networking	8 基础 Basic			Z Zigbee
		9 待定义 To be defined	9 高空 High Altitude			

第三部分 / Chapter 3

使用指南 / User Guide

3.1 安装指南 / Installation Guide

3.1.1 安装位置说明 / Installation Position

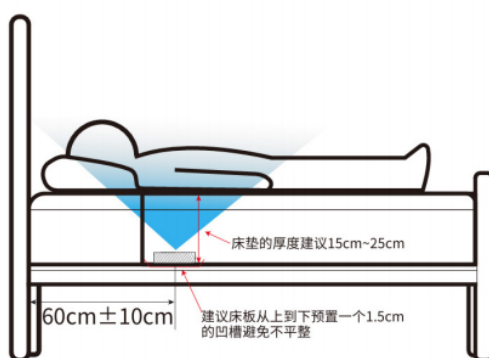


图 3.1: 安装位置示意图
Fig3.1: Installation Position Diagram

传感器应放置在床垫下方、床板上方，人体平躺时，胸腔应位于传感器正上方。建议床垫厚度在 15~25 cm 的范围。

The sensor should be placed between the mattress and the bed board. When a person is lying flat, the chest should be positioned directly above the sensor. A mattress thickness of 15~25 cm is recommended.

3.1.2 供电与电源要求 / Power Supply Requirements

- 使用 5V2A USB Type-A 接口的适配器进行供电。
Use a 5V2A USB Type-A adapter for power supply.
- 过大的电源纹波可能对雷达传感器造成干扰产生误报，建议供电电源纹波应低于 100mV。
Excessive power supply ripple may cause interference to the radar sensor and result in false alarms. It is recommended that the power supply ripple should be less than 100mV.

3.1.3 安装注意事项 / Installation Precautions

- 雷达安装位置应避免受周边震动源影响，包括但不限于：冰箱、空调外机、水泵、发电机等。
The radar installation location should avoid being affected by nearby vibration sources, including but not limited to: refrigerators, air conditioner outdoor units, water pumps, generators, etc.
- 雷达感应范围内避免有处于晃动、运动、明显振动的物体及大面积金属，包括但不限于：吊灯、风扇、空调、绿植、窗帘及有金属涂层的镜子等。

Avoid objects that are swaying, moving, or significantly vibrating, as well as large metal surfaces within the radar detection range, including but not limited to: chandeliers, fans, air conditioners, plants, curtains, and mirrors with metal coatings.

- 雷达传感器安装时，背面如有铝基板或其他金属板，应抬高一定高度，与金属平面保持 5mm 以上距离，不能紧贴或接触金属平面；雷达传感器天线前方不能有金属遮挡以及大电流电缆覆盖，避免正对驱动电源，同时尽量远离驱动电源的整流桥、变压器、开关管等大功率器件。

When installing the radar sensor, if there is an aluminum substrate or other metal plate on the back, it should be raised at least 5mm away from the metal surface and must not be in direct contact. The front of the radar sensor antenna should not be blocked by metal or covered by high-current cables. Avoid facing the drive power supply directly, and keep away from high-power components such as rectifier bridges, transformers, and switching transistors.

- 雷达传感器对塑胶、木质材料穿透性效果较好，对金属或有金属涂层的材质则不能穿透，如用户产品的外壳为玻璃、陶瓷、碳纤维等特殊材料时，请以实测效果为准，必要时请联系易探科技技术人员做适用性调试。

The radar sensor has good penetration through plastic and wood materials, but cannot penetrate metal or metal-coated materials. If the product housing is made of special materials such as glass, ceramic, or carbon fiber, please refer to actual test results and contact EasyDetek technical personnel for applicability debugging if necessary.

- 为了达到较好的存在检测效果，提升使用体验，建议：探测覆盖空间内避免存在可动物体，包括但不限于飘动的窗帘，晃动的绿植、转动的风扇等。

To achieve better presence detection and improve user experience, it is recommended to avoid movable objects within the detection coverage area, including but not limited to: fluttering curtains, swaying plants, rotating fans, etc.

- 如雷达传感器与无线通讯模块（NB、蓝牙、WIFI、2.4G 模块）等一起使用时，应拉开间距。推荐安装时与路由器、无线热点等大功率无线通讯设备保持 1m 以上间距。

When the radar sensor is used together with wireless communication modules (NB, Bluetooth, WiFi, 2.4G modules), maintain sufficient spacing. It is recommended to keep a distance of at least 1m from high-power wireless communication devices such as routers and wireless hotspots during installation.

3.2 产品功能 / Product Functions

实时上报以下状态，并支持统计分析。

The following statuses are reported in real time and support statistical analysis.

- **睡眠统计分析值 / Sleep Statistics:** 包含入睡时间、起床时间、睡眠时长、睡眠质量分数、浅睡/深睡/REM 占比、翻身次数。

Includes sleep onset time, wake-up time, sleep duration, sleep quality score, light sleep/deep sleep/REM ratio, and turning count.

- **体征实时状态值 / Vital Signs Status:** 包含在离位/睡眠状态、体动数值、心率、呼吸率、心率状态、呼吸率状态。

Includes presence/sleep status, body movement value, heart rate, respiration rate, heart rate status, and respiration rate status.

设备配备红绿双色指示灯，指示当前工作状态。

The device is equipped with red and green LED indicators to indicate the current working status.

表 3.1: 指示灯状态表
Tab3.1: LED Indicator Status

指示灯状态 LED Status	说明 Description
绿色常亮 Green Constant On	正在进行上电初始化 Power-on initialization
绿色单次闪烁 Green Single Flash	上报体征检测数据 Reporting vital signs data
红色单次闪烁 Red Single Flash	发送设备在线通讯心跳 Sending heartbeat
绿色连续快速闪烁 Green Rapid Flash	正在进行 OTA 升级 OTA upgrading
红色短亮 Red Brief On	OTA 升级失败 OTA upgrade failed
红色闪烁 Red Flashing	联网失败 Network connection failed
红色常亮 Red Constant On	SIM 卡异常或设备硬件异常 SIM card or hardware error

3.3 客户端使用指南 / Client User Guide

3.3.1 登录客户端查看数据 / Login to Client to View Data

网页端查看体征检测数据：进入网页
<https://easydetekiot.com/login>
输入用户名和密码登录。

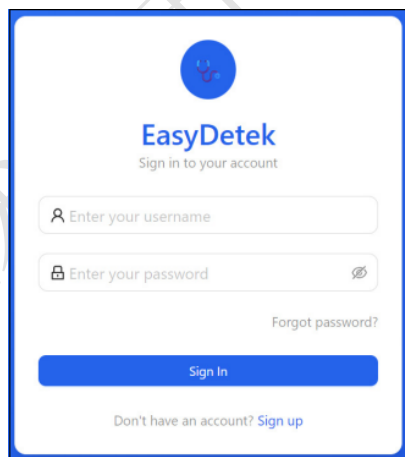


图 3.2: 网页端登录页面
Fig3.2: Web Login Page

进入小程序右下角的【个人中心】，点击【登录】。若还没有账号，可点击登录页下方的【立即注册】进入注册页面。注册页面支持两种注册方式：通过微信授权手机号码一键注册，或输入手机号码获取验证码注册。填完表单后，点击页面下方的【注册】按钮即可完成注册。



图 3.3: 小程序端登录页面
Fig3.3: Mini Program Login Page

3.3.2 查看与绑定设备 / View and Bind Devices

查看我的设备 / View My Devices

登录成功后可以看到账号的用户信息，点击导航栏【我的设备】即可看到该账号下绑定的所有设备。
After logging in, you can see the account user information. Click [My Devices] in the navigation bar to view all devices bound to the account.

绑定我的设备 / Bind My Device

在【我的设备】页面中，可以点击右上角的添加按钮，添加（绑定）新的设备。点击任意设备卡片即可进入对应的设备详情页，在设备详情页面中可以查看实时监测数据、历史数据、睡眠质量报告等内容。以上为小程序端的查看流程，网页端登录后可直接查看该账号下绑定的所有设备。

On the [My Devices] page, you can click the add button in the upper right corner to add (bind) a new device. Click any device card to enter the corresponding device details page, where you can view real-time monitoring data, historical data, sleep quality reports, etc. The above is the viewing process in the mini program; on the web portal, you can directly view all devices bound to the account after logging in.

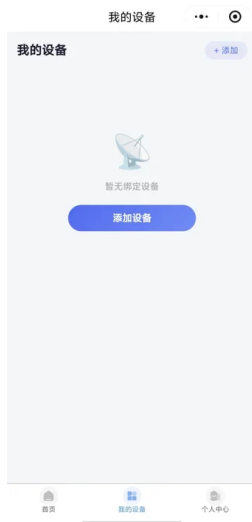


图 3.4: 查看我的所有设备
Fig3.4: View All My Devices



图 3.5: 绑定我的设备
Fig3.5: Bind My Device

3.3.3 设备详情页 / Device Details Page

设备详情页面中，可以通过点击图表右上角的【1h / 6h / 12h / 24h】按钮，分别查看各个时间段的数据；睡眠质量报告则可以通过点击右上角的【当天 / 本周 / 本月】按钮，查看不同时间段的汇总报告。

On the device details page, click the [1h / 6h / 12h / 24h] buttons at the top right of the chart to view data of different time periods; the sleep quality report can be viewed by clicking the [Today / This Week / This Month] buttons at the top right to view summary reports of different time periods.



图 3.6: 小程序数据详情页面
Fig3.6: Mini Program Data Details Page



图 3.7: 小程序睡眠质量报告
Fig3.7: Mini Program Sleep Quality Report



图 3.8: 网页端详细数据
Fig3.8: Web Detailed Data

3.3.4 网页端设置配置参数 / Web Configuration Settings

点击设备卡片上的三个点符号再点击设置或者在设备详情页面点击齿轮图标可以进入设备设置界面。
Click the three-dot symbol on the device card and then select Settings, or click the gear icon on the device details page to enter the device settings interface.

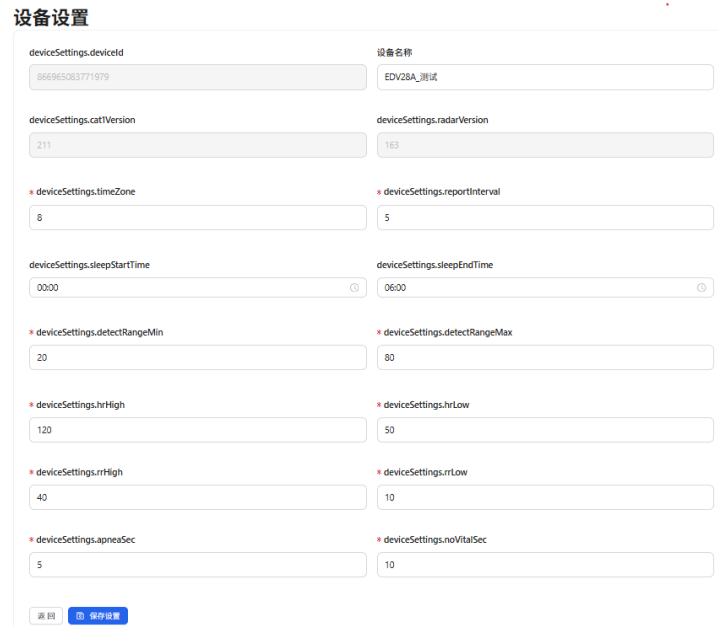


图 3.9: 设备设置界面
Fig3.9: Device Settings Interface

设置界面可以查看设备当前的软件版本信息、设置睡眠和体征检测预警的相关配置参数。

为了提高睡眠质量报告的准确性，需要设置时区、日常睡眠的一般起始时间和一般结束时间。每天的睡眠质量报告生成于设定睡眠结束时间与实际睡眠结束时间两者中最晚的时间。

检测的距离范围建议保持默认值，如果需要修改请咨询技术支持人员。

确认修改配置参数后请点击保存设置按钮。

The settings interface allows you to view the current software version information of the device and configure parameters related to sleep and vital signs detection alerts.

To improve the accuracy of the sleep quality report, you need to set the time zone, the usual start time and end time of daily sleep. The daily sleep quality report is generated at the later of the set sleep end time and the actual sleep end time.

It is recommended to keep the detection distance range at the default value. If modification is required, please consult technical support personnel.

After confirming the configuration parameter changes, please click the Save Settings button.

3.3.5 体征实时状态与统计分析值 / Vital Signs Real-Time Status and Statistics

1. 体征实时状态值 / Vital Signs Real-Time Status:

- **在位/离位状态:** 当人进入检测区域后，即会显示在位，人离位一段时间后，会显示离位。
Presence/Absence Status: When a person enters the detection area, it will display "present"; after the person leaves for a while, it will display "absent".
- **清醒/睡眠状态:** 传感器检测到人睡眠后，分别会进入眼动、浅睡、深睡状态，并进行相应切换。
Awake/Sleep Status: After the sensor detects that the person is asleep, it enters REM, light sleep, and deep sleep states respectively and switches accordingly.
- **呼吸和心率值:** 当人进入检测区域后，不会立即显示呼吸和心率值，需等待被测对象处于静息状态后，才

能显示呼吸和心率值。

Respiration & Heart Rate Values: After a person enters the detection area, respiration and heart rate values are not displayed immediately; they are only displayed after the subject is in a resting state.

2. 体征和睡眠统计分析值 / Vital Signs and Sleep Statistics:

- 从人体静息后，最快 10s 可以输出体征值；人体未处于静息状态，传感器可能会一直无法输出有效的体征值，直到用户处于静息状态。

Vital sign values can be output within as fast as 10 seconds after the person rests. If the person is not in a resting state, the sensor may not output valid vital sign values until the user is at rest.

第四部分 / Chapter 4

协议规范 / Protocol Specification

4.1 通信接口 / Communication Interface

本产品采用 USB 接口 (Type-C) 进行供电和数据通信。通过 USB 连接线接入 5V 直流电源适配器为设备供电。

This product uses a USB interface (Type-C) for power supply and data communication. Connect the device to a 5V DC power adapter via a USB cable for power supply.

4.2 通讯协议 / Communication Protocol

本产品由雷达监测设备、云端服务平台与客户端 (App/网页) 构成, 通信分为设备接入与应用接口两个层级。

This product consists of a radar monitoring device, a cloud service platform, and clients (App/Web). The communication system is divided into two layers: **device access** and **application interface**.

4.2.1 设备接入协议 / Device Access Protocol (MQTT)

设备通过 MQTT 协议与云端建立长连接, 完成数据上报、远程配置与固件升级 (OTA)。

The device establishes a persistent connection with the cloud via the MQTT protocol for data reporting, remote configuration, and firmware upgrade (OTA).

表 4.1: MQTT 参数表
Tab4.1: MQTT Parameters

项目 / Item	说明 / Description
传输协议 / Protocol	MQTT 3.1.1 / 5.0
服务器 / Server	www.easydetekiot.com
端口 / Port	1883 (TCP)
鉴权 / Authentication	用户名 / 密码
保活周期 / Keep Alive	60 s
通信模型 / Model	发布/订阅 (Topic)

Topic 规划 / Topic Planning:

- 数据上报 / Data Report: radar/{device_type}/{device_id}/data

样例 / Example: Topic:radar/edv28a/866965083771979/data, QoS: 0

```
{  
  "time_stamp": 1786333856,  
  "breath_state": 0,  
  "breath_confidence": 2,  
  "distance": 29,  
  "breath_rate": 15,
```

```
"sleep_state": 1,  
"heart_confidence": 1,  
"heart_rate": 52,  
"heart_state": 0,  
"movement": 10  
}
```

- 配置下发 / Config Set: radar/{device_type}/{device_id}/config/set

样例 / Example: Topic:radar/edv28a/866965083771979/config/set QoS: 0

```
{  
  "apnea_sec": 5,  
  "detect_range_max": 80,  
  "detect_range_min": 20,  
  "device_id": "866965083771979",  
  "device_name": "EDV28A_ 测试",  
  "hr_high": 120,  
  "hr_low": 50,  
  "interest_sleep_end_time": 360,  
  "interest_sleep_start_time": 0,  
  "no_vital_sec": 10,  
  "report_interval": 5,  
  "rr_high": 40,  
  "rr_low": 10,  
  "time_zone": 8  
}
```

- 配置确认 / Config Ack: radar/{device_type}/{device_id}/config/ack

样例 / Example: Topic:radar/edv28a/866965083771979/config/ack, QoS: 0

```
{  
  "result": "success",  
  "msg": "Config updated"  
}
```

- 固件请求 / OTA Request: radar/{device_type}/{device_id}/getOta

样例 / Example: Topic:radar/edv28a/866965083771979/getOta QoS: 0

```
{  
  "deviceType": "EDV28A",  
  "moduleType": "Air780EHV"  
}
```

- 固件响应 / OTA Response: radar/{device_type}/{device_id}/otaRes

样例 / Example: Topic:radar/edv28a/866965083771979/otaRes QoS: 0

```
{  
  "filename": "EDV28A-Air780EHV-v2.0.7-ota-0xF3620EF1.bin",  
  "ossUrl": "http://120.24.115.60:8080/firmware-files/EDV28A/Air780EHV/  
EDV28A-Air780EHV-v2.0.7-ota-0xF3620EF1.bin",  
  "status": "succse"  
}
```

4.2.2 应用接口协议 / Application Interface Protocol (REST/HTTPS)

客户端基于 HTTPS 的 RESTful API 与云端交互，提供用户管理、设备绑定、数据查询与配置管理。
Clients interact with the cloud via HTTPS-based RESTful APIs for user management, device binding, data query, and configuration management.

表 4.2: REST API 参数表
Tab4.2: REST API Parameters

项目 / Item	说明 / Description
协议 / Protocol	HTTPS (REST)
Base URL	http://www.easydetekiot.com:3000/api
数据格式 / Data Format	application/json
鉴权 / Authentication	Bearer Token (登录后下发)
响应结构 / Response	统一 ApiResponse<T>: code / msg / data / timestamp

4.2.3 安全机制 / Security Mechanisms

- 传输加密：设备—云端、客户端—云端通信均经加密通道传输。

Transport Encryption: Communication between device and cloud, as well as between client and cloud, is transmitted through encrypted channels.

- 身份认证：设备采用账号密码鉴权；客户端采用 Bearer Token 鉴权，密码经 RSA 加密后传输。

Authentication: Devices use username/password authentication; clients use Bearer Token authentication, with passwords encrypted via RSA for transmission.

- 数据隔离：设备与用户账号绑定，数据按账号隔离。

Data Isolation: Devices are bound to user accounts, and data is isolated by account.

第五部分 / Chapter 5

保障条款 / Warranty Terms

5.1 安全合规 / Safety Compliance

1. 射频安装规范

本设备搭载毫米波雷达模组，请勿私自拆解设备壳体；安装时传感器背部避免紧贴大面积金属板材，雷达探测前方不宜存在固定金属遮挡，防止影响探测效果。

RF Installation Specifications: This device is equipped with a millimeter-wave radar module. Do not disassemble the device housing without authorization. During installation, avoid placing the sensor back tightly against large metal surfaces. There should be no fixed metal obstructions in front of the radar detection area to prevent affecting detection performance.

2. 线缆使用规范

请勿大力拉扯、反复弯折设备自带连接线，避免线缆内部受损。

Cable Usage Specifications: Do not pull or repeatedly bend the device's connecting cable to avoid internal damage.

3. 环境使用要求

设备通电运行、长期存放需满足产品规定温度范围，避免长期放置在潮湿、具有腐蚀性的环境中。

Environmental Requirements: The device must be operated and stored within the specified temperature range. Avoid prolonged exposure to humid or corrosive environments.

4. 安装布置规范

安装点位远离水泵、风扇、大功率变频设备、无线路由器等电磁干扰源；探测视野内避免存在持续晃动障碍物。

Installation Layout Specifications: Keep the installation site away from electromagnetic interference sources such as water pumps, fans, high-power frequency converters, and wireless routers. Avoid persistent moving obstructions within the detection field of view.

5. 使用禁令

禁止私自改装设备硬件；禁止人为持续遮挡雷达探测区域；禁止将设备用于法律法规不允许的信息采集场景。

Usage Prohibitions: Do not modify the device hardware without authorization. Do not intentionally block the radar detection area. Do not use the device for information collection scenarios prohibited by laws and regulations.

6. 报废处置

设备报废请勿随意丢弃，请按照当地电子废弃物回收相关规定处理。

Disposal: Do not discard the device arbitrarily at the end of its service life. Please dispose of it in accordance with local electronic waste recycling regulations.

5.2 免责声明 / Disclaimers

1. 违规使用免责

若未遵循 5.1 安全合规要求安装、使用设备，由此造成设备损坏、探测功能异常，设备将丧失质保权益，本公司不承担相应损失。

Disclaimer for Improper Use: If the device is installed or used without following the safety compliance requirements in Section 5.1, resulting in device damage or abnormal detection function, the warranty shall be voided, and the company shall not be liable for any resulting losses.

2. 检测局限性免责

心率、呼吸、人体在离位、睡眠状态依靠胸腔微动识别，受床垫条件、被褥厚度、多人同床、外部震动干扰影响，监测数据出现合理误差属于正常情况。

Disclaimer for Detection Limitations: Heart rate, respiration, presence/absence, and sleep status are identified through chest micro-movements. Reasonable deviations in monitoring data due to mattress conditions, blanket thickness, multiple persons in bed, or external vibration interference are considered normal.

3. 数据用途免责

设备输出各类体征、睡眠统计数据仅可作为日常康养参考，不可作为专业医疗诊断依据，不能单独依靠设备监测结果判断身体健康状况。

Disclaimer for Data Usage: All vital signs and sleep statistics output by the device are for daily health reference only. They shall not be used as a basis for professional medical diagnosis, and health conditions should not be determined solely based on the device monitoring results.

4. 网络云端免责

小程序、公众号平台的数据展示依赖移动通信网络与云端服务，因网络故障、服务器维护造成的数据延迟、数据丢失，本公司不予承担责任。

Disclaimer for Network and Cloud Services: Data display on the Mini Program and Official Account platforms relies on mobile communication networks and cloud services. The company shall not be liable for data delays or loss caused by network failures or server maintenance.

5. 外部不可抗力免责

雷击、电网异常浪涌、周边强电磁干扰、自然灾害等外部因素导致设备故障，不在保修责任范围。

Disclaimer for Force Majeure: Device failures caused by external factors such as lightning strikes, abnormal power surges, strong electromagnetic interference, or natural disasters are not covered by the warranty.

6. 人为损坏免责

进水、外力撞击挤压、私自拆机改造、非官方维修造成的设备损坏，不在保修范围内。

Disclaimer for Human-Induced Damage: Device damage caused by water ingress, external impact or extrusion, unauthorized disassembly or modification, or non-official repairs is not covered by the warranty.

7. 版本迭代说明

产品软硬件持续迭代优化，本说明书内容仅供参考，功能调整更新不再另行通知。

Version Iteration Note: The product hardware and software are continuously iterated and optimized. This manual is for reference only. Function adjustments and updates will not be separately notified.

5.3 售后服务 / After-Sales Service



图 5.1: 微信公众号
Fig5.1: WeChat Official Account

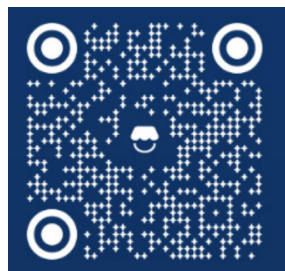


图 5.2: 微信小店
Fig5.2: WeChat Store